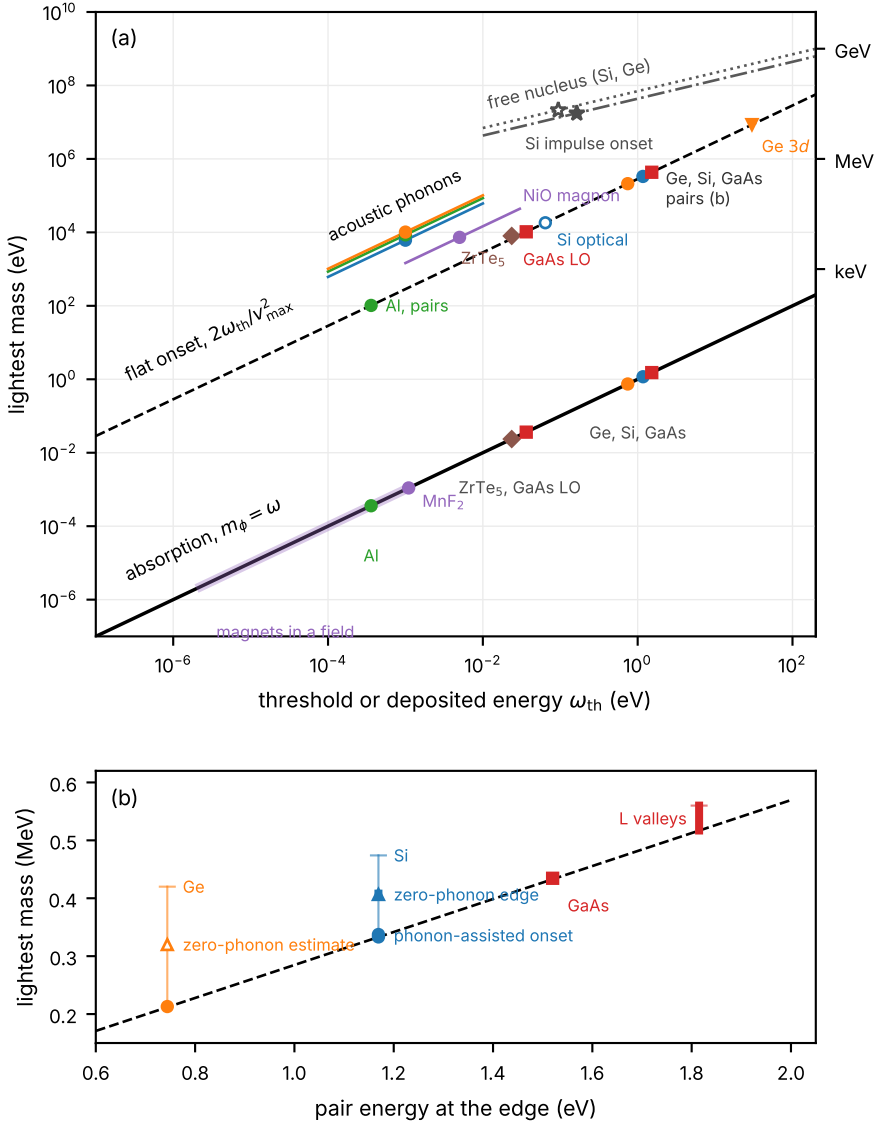
(a)
free nucleus (Si, Ge)
Si impulse onset
Ge 3d
GeV
MeV
keV
acoustic phonons
NiO magnon
Ge, Si, GaAs
pairs (b)
Si optical
GaAs LO
ZrTe5
Al, pairs
flat onset, 2ωth/v²max
Ge, Si, GaAs
ZrTe5, GaAs LO
MnF2
absorption, mϕ = ω
Al
magnets in a field
lightest mass (eV)
threshold or deposited energy ωth (eV)
(b)
L valleys
Si
Ge
zero-phonon edge
GaAs
zero-phonon estimate
phonon-assisted onset
lightest mass (MeV)
pair energy at the edge (eV)